

Transportation problem

$$Z = 2x_1 + 3x_2 \rightarrow \min$$

$$\begin{cases} 4x_1 - x_2 \geq 6 \\ 9x_1 + 8x_2 \leq 62 \\ -3x_1 + 11x_2 \geq 16 \\ x_1, x_2 \geq 0 \end{cases}$$

1) $4x_1 - x_2 = 6$

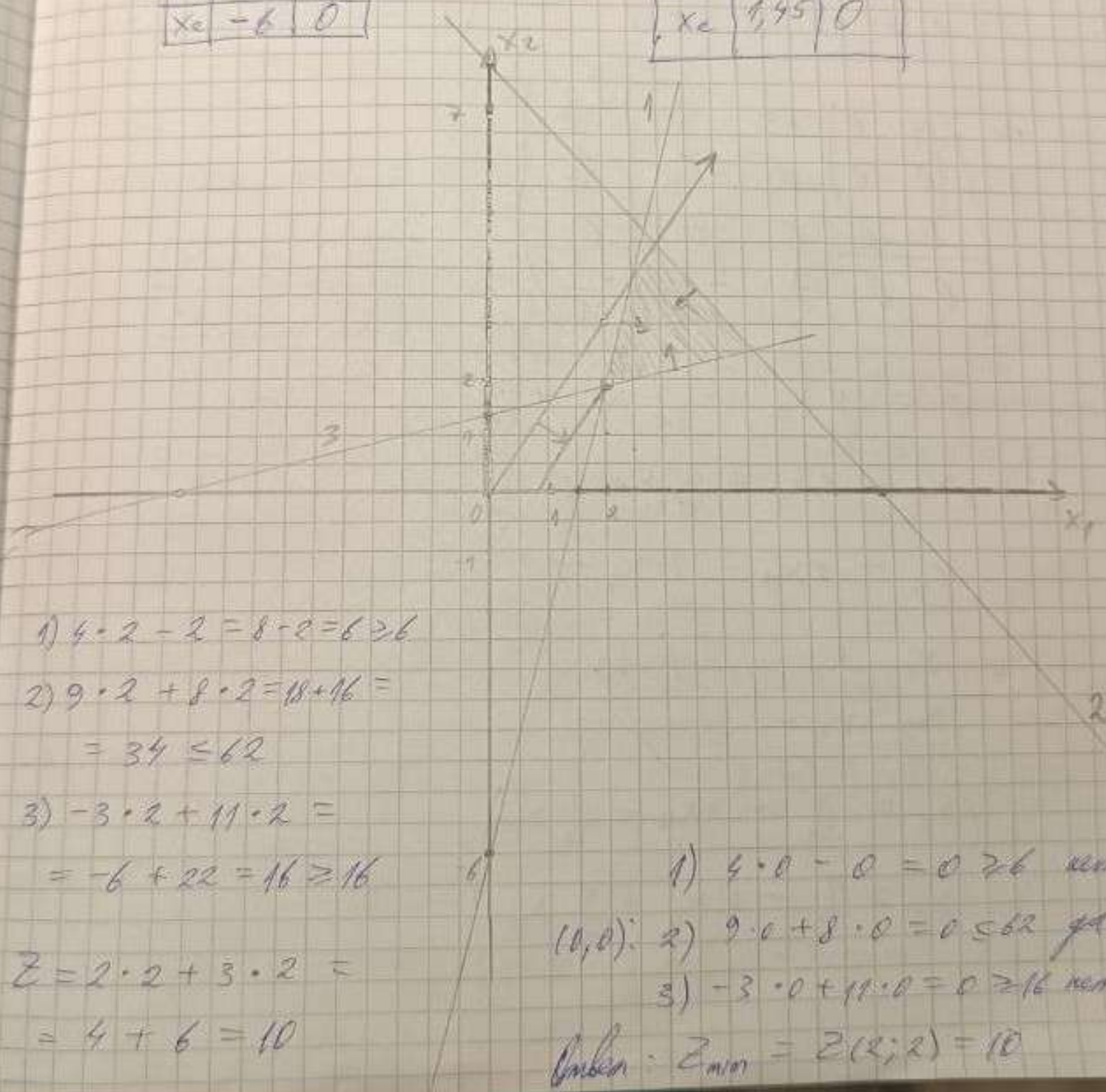
x_1	0	1,5
x_2	-6	0

2) $9x_1 + 8x_2 = 62$

x_1	0	6,89
x_2	7,75	0

3) $-3x_1 + 11x_2 = 16$

x_1	0	-5,33
x_2	1,45	0



1) $4 \cdot 2 - 2 = 8 - 2 = 6 \geq 6$

2) $9 \cdot 2 + 8 \cdot 2 = 18 + 16 = 34 \leq 62$

3) $-3 \cdot 2 + 11 \cdot 2 = -6 + 22 = 16 \geq 16$

$Z = 2 \cdot 2 + 3 \cdot 2 = 4 + 6 = 10$

1) $4 \cdot 0 - 0 = 0 \geq 6$ nem

(0,0): 2) $9 \cdot 0 + 8 \cdot 0 = 0 \leq 62$ ja

3) $-3 \cdot 0 + 11 \cdot 0 = 0 \geq 16$ nem

Összesen: $Z_{\min} = Z(2; 2) = 10$

$$Z = x_1 - 2x_2 + 2x_3 + x_4 \rightarrow \min$$

$$\begin{cases} x_1 + x_2 - x_3 = 2 \\ x_1 + 3x_2 + 3x_3 - x_4 = 3 \\ x_1 - x_2 - 5x_3 + x_4 = 1 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases}$$

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 0 & 2 \\ 1 & 3 & 3 & -1 & 3 \\ 1 & -1 & -5 & 1 & 1 \\ 1 & -2 & 2 & 1 & 0 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} 1 & 1 & -1 & 0 & 2 \\ 0 & 2 & 4 & -1 & 1 \\ 0 & -2 & -4 & 1 & -1 \\ 0 & -3 & 3 & 1 & -2 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & -3 & \frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 2 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 9 & -\frac{1}{2} & -\frac{1}{2} \end{array} \right)$$

$$Z = 9x_3 - \frac{1}{2}x_4 + \frac{1}{2}$$

$$x_1 - 3x_3 + \frac{1}{2}x_4 = \frac{3}{2}$$

$$x_2 + 2x_3 - \frac{1}{2}x_4 = \frac{1}{2}$$

x_1, x_2 — базисные

x_3, x_4 — свободные

$$\begin{cases} x_1 = \frac{3}{2} + 3x_3 - \frac{1}{2}x_4 \\ x_2 = \frac{1}{2} - 2x_3 + \frac{1}{2}x_4 \end{cases}$$

$$x_1 \geq 0 \Rightarrow \frac{3}{2} + 3x_3 - \frac{1}{2}x_4 \geq 0$$

$$x_2 \geq 0 \Rightarrow \frac{1}{2} - 2x_3 + \frac{1}{2}x_4 \geq 0 \Rightarrow$$

$$\Rightarrow \begin{cases} -3x_3 + \frac{1}{2}x_4 \leq \frac{3}{2} \\ 2x_3 - \frac{1}{2}x_4 \leq \frac{1}{2} \\ x_3, x_4 \geq 0 \end{cases} \cdot 2 =$$

$$\Rightarrow \begin{cases} -6x_3 + x_4 \leq 3 \\ 4x_3 - x_4 \leq 1 \\ x_3, x_4 \geq 0 \end{cases}$$

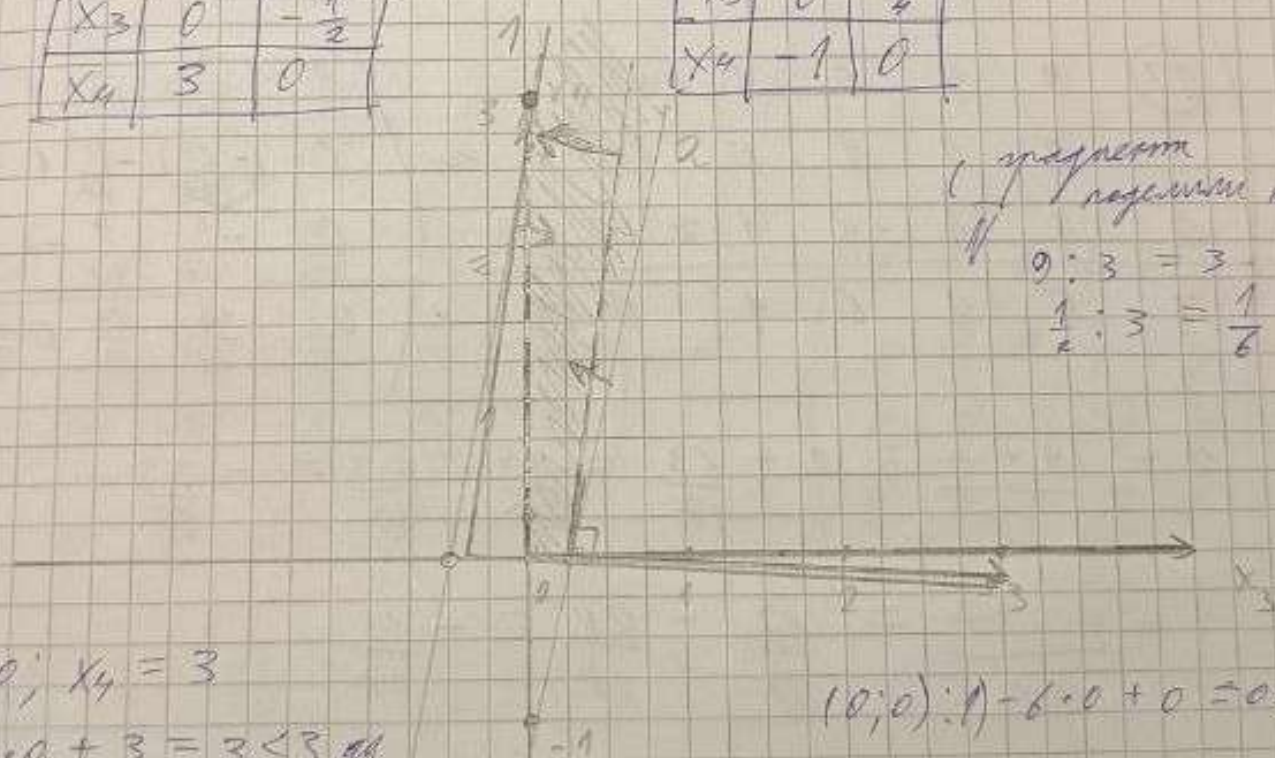
$$Z = 9x_3 - \frac{1}{2}x_4 + \frac{1}{2}$$

$$1) -6x_3 + x_4 = 3$$

$$2) 4x_3 - x_4 = 1$$

x_3	0	$-\frac{1}{2}$
x_4	3	0

x_3	0	$\frac{1}{4}$
x_4	-1	0



проверим
поделим на 3
// $0 : 3 = 3$
 $\frac{1}{2} : 3 = \frac{1}{6}$

$$x_3 = 0, x_4 = 3$$

$$1) -6 \cdot 0 + 3 = 3 \leq 3 \text{ да}$$

$$2) 4 \cdot 0 - 3 = -3 \leq 1 \text{ да}$$

$$(0; 0): 1) -6 \cdot 0 + 0 = 0 \leq 3 \text{ да}$$

$$2) 4 \cdot 0 - 0 = 0 \leq 1 \text{ да}$$

$$Z = 9 \cdot 0 - \frac{1}{2} \cdot 3 + \frac{1}{2} = 0 - \frac{3}{2} + \frac{1}{2} = -1$$

$$x_1 = \frac{3}{2} + 3 \cdot 0 - \frac{1}{2} \cdot 3 = \frac{3}{2} - \frac{3}{2} = 0 \geq 0$$

$$x_2 = \frac{1}{2} - 2 \cdot 0 + \frac{1}{2} \cdot 3 = \frac{1}{2} + \frac{3}{2} = 2 \quad \text{? 0 ?}$$

Optimal: $Z_{\text{opt}} = Z(0; 2; 0; 3) = -1$

17

$$Z = 3x_1 + 5x_2 + 2x_3 + x_5 \rightarrow \max$$

$$\begin{cases} 17x_1 + 9x_2 + 4x_3 + x_5 = 51 \\ 7x_1 + 2x_2 + x_3 + x_4 = 12 \\ 2x_1 + x_2 + x_3 - x_4 + x_5 = 15 \\ x_1, x_2, x_3, x_4, x_5 \geq 0 \end{cases}$$

~~$$\begin{array}{ccccc|c} 17 & 9 & 4 & 0 & 1 & 51 \\ 7 & 2 & 1 & 1 & 0 & 12 \\ 2 & 1 & 1 & -1 & 1 & 15 \end{array}$$~~

$$\begin{pmatrix} \textcircled{17} & 9 & 4 & 0 & 1 & 51 \\ 7 & 2 & 1 & 1 & 0 & 12 \\ 2 & 1 & 1 & -1 & 1 & 15 \\ 3 & 5 & 2 & 0 & 1 & 0 \end{pmatrix} \sim \begin{pmatrix} 1 & \frac{9}{17} & \frac{4}{17} & 0 & \frac{1}{17} & \frac{51}{17} \\ 0 & \textcircled{-\frac{29}{17}} & -\frac{11}{17} & 1 & -\frac{7}{17} & -9 \\ 0 & -\frac{1}{17} & \frac{9}{17} & -1 & \frac{15}{17} & 9 \\ 0 & \frac{40}{17} & \frac{22}{17} & 0 & \frac{14}{17} & 9 \end{pmatrix}$$

$$2 - \frac{9 \cdot 7}{17} = 2 - \frac{63}{17} = \frac{34}{17} - \frac{63}{17} = -\frac{29}{17}$$

$$1 - \frac{4 \cdot 7}{17} = \frac{17}{17} - \frac{28}{17} = -\frac{11}{17}$$

$$1 - 0 \cdot \frac{7}{17} = 1 - 0 = 1$$

$$0 - \frac{1 \cdot 7}{17} = -\frac{7}{17}$$

$$\begin{aligned} 12 - \frac{51 \cdot 7}{17} &= 12 - \frac{357}{17} \\ &= 12 - 21 = -9 \end{aligned}$$

$$1 - \frac{2 \cdot 2}{17} = \frac{17}{17} - \frac{16}{17} = \frac{1}{17}$$

$$1 - \frac{4 \cdot 2}{17} = \frac{17}{17} - \frac{8}{17} = \frac{9}{17}$$

$$-1 - \frac{0 \cdot 2}{17} = -1$$

$$1 - \frac{1 \cdot 2}{17} = \frac{17}{17} - \frac{2}{17} = \frac{15}{17}$$

$$15 - \frac{51 \cdot 2}{17} = 15 - \frac{102}{17} = 15 - 6 = 9$$

$$5 - \frac{9 \cdot 3}{17} = 5 - \frac{27}{17} = \frac{25}{17} - \frac{27}{17} = \frac{48}{17}$$

$$2 - \frac{4 \cdot 3}{17} = \frac{34}{17} - \frac{12}{17} = \frac{22}{17}$$

$$1 - \frac{1 \cdot 3}{17} = \frac{17}{17} - \frac{3}{17} = \frac{14}{17}$$

$$0 - \frac{51 \cdot 3}{17} = \frac{153}{17} = 9$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & \frac{1}{29} & \frac{9}{29} & -\frac{2}{29} & \frac{6}{29} \\ 0 & 1 & \frac{11}{29} & -\frac{17}{29} & \frac{7}{29} & \frac{153}{29} \\ 0 & 0 & -\frac{16}{29} & -\frac{30}{29} & \frac{26}{29} & \frac{240}{29} \\ 0 & 0 & 0 & 2 & 0 & -27 \end{array} \right)$$

$$\sim \left(\begin{array}{cccc|c} 1 & 0 & 0 & \frac{3}{8} & -\frac{1}{8} & -\frac{3}{8} \\ 0 & 1 & 0 & \frac{1}{8} & -\frac{3}{8} & -\frac{9}{8} \\ 0 & 0 & 1 & -\frac{15}{8} & \frac{13}{8} & \frac{135}{8} \\ 0 & 0 & 0 & 2 & 0 & -27 \end{array} \right)$$

$$-\frac{11}{17} : -\frac{29}{17} = \frac{-11}{17} \cdot \frac{17}{-29} = \frac{11}{29}$$

$$-\frac{17}{17} : -\frac{29}{17} = -\frac{17}{29}$$

$$-\frac{7}{17} : -\frac{29}{17} = \frac{7}{29}$$

$$-9 : -\frac{29}{17} =$$

$$= -9 \cdot \left(\frac{17}{29} \right) = \frac{153}{29}$$

$$\frac{4}{17} = \frac{\frac{9}{17} \cdot \left(1 - \frac{11}{17}\right)}{-\frac{29}{17}} = \frac{4}{17} - \left(\frac{99}{17} \cdot \frac{17}{29}\right) =$$

$$= \frac{4}{17} - \frac{99}{29} =$$

$$Z = 2x_4 + 27$$

x_1, x_2, x_3 — free.

$$x_1 + \frac{3}{8}x_4 - \frac{1}{8}x_5 = -\frac{3}{8}$$

$$x_2 + \frac{1}{8}x_4 - \frac{3}{8}x_5 = \frac{9}{8}$$

$$x_3 - \frac{15}{8}x_4 + \frac{13}{8}x_5 = \frac{135}{8}$$

$$x_1 = -\frac{3}{8} - \frac{3}{8}x_4 + \frac{1}{8}x_5$$

$$x_2 = \frac{9}{8} - \frac{1}{8}x_4 + \frac{3}{8}x_5$$

$$x_3 = \frac{135}{8} + \frac{15}{8}x_4 - \frac{13}{8}x_5$$

$$x_1, x_2, x_3 \geq 0 \Rightarrow$$

$$\Rightarrow -\frac{3}{8} - \frac{3}{8}x_4 + \frac{1}{8}x_5 \geq 0$$

$$-\frac{9}{8} - \frac{1}{8}x_4 + \frac{3}{8}x_5 \geq 0$$

$$\frac{135}{8} + \frac{15}{8}x_4 - \frac{13}{8}x_5 \geq 0$$

$$Z = 2x_4 + 27 \rightarrow \max$$

$$\begin{cases} \frac{3}{8}x_4 - \frac{1}{8}x_5 \leq -\frac{3}{8} \\ \frac{1}{8}x_4 - \frac{3}{8}x_5 \leq -\frac{9}{8} \\ -\frac{15}{8}x_4 + \frac{13}{8}x_5 \leq \frac{135}{8} \\ x_4, x_5 \geq 0 \end{cases}$$

$$1) \frac{3}{8} x_4 - \frac{1}{8} x_5 = -\frac{3}{8}$$

x_4	0	-1
x_5	3	0

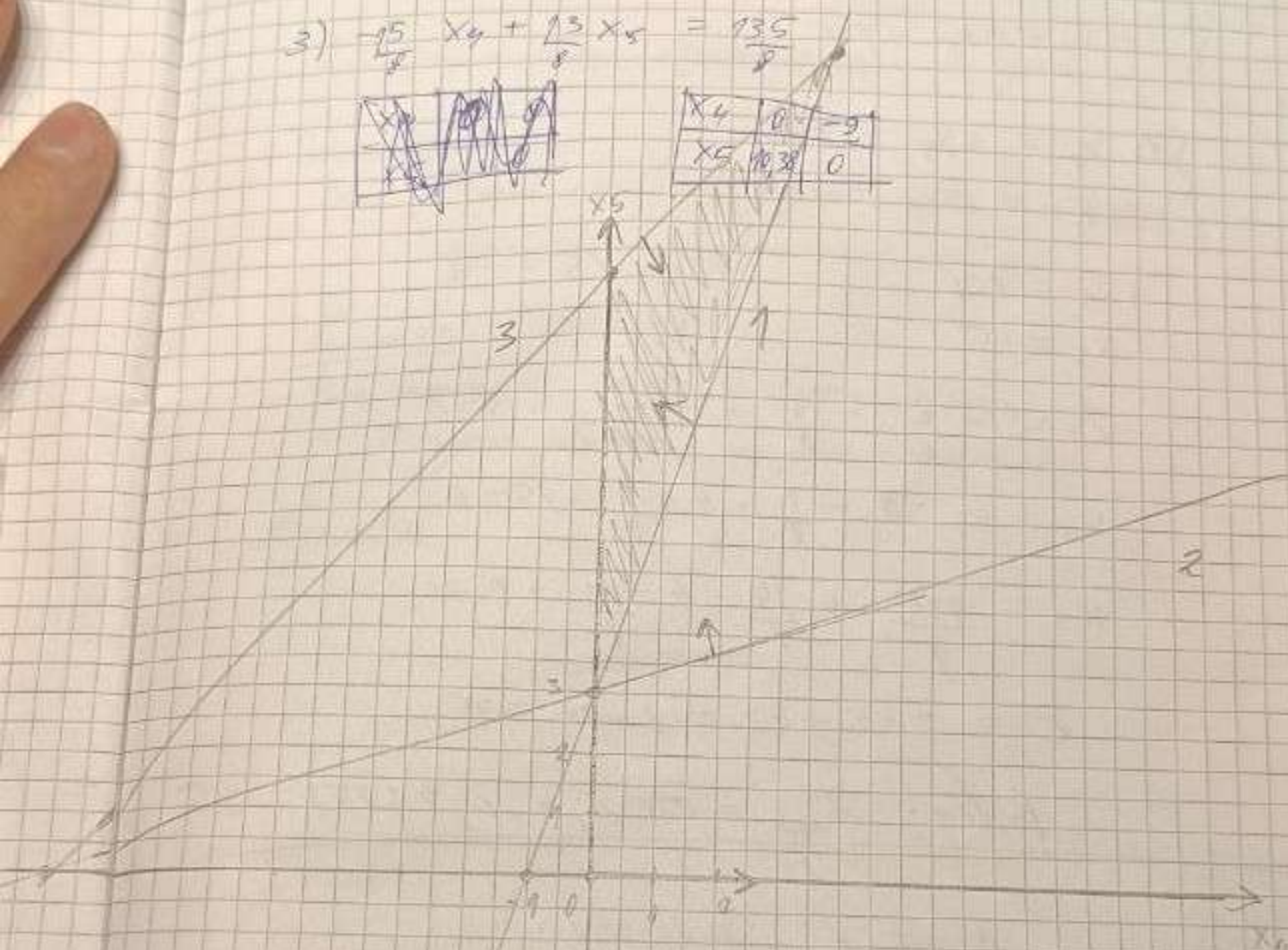
$$2) \frac{1}{8} x_4 - \frac{3}{8} x_5 = -\frac{9}{8}$$

x_4	0	-9
x_5	3	0

$$3) -\frac{15}{8} x_4 + \frac{13}{8} x_5 = \frac{135}{8}$$

x_4	0	9
x_5	10,38	0

x_4	0	-9
x_5	10,38	0



$$(0,0): \frac{3}{8} \cdot 0 - \frac{1}{8} \cdot 0 = 0 \leq -\frac{3}{8} \quad \text{nicht}$$

$$\frac{1}{8} \cdot 0 - \frac{3}{8} \cdot 0 = 0 \leq -\frac{9}{8} \quad \text{nicht}$$

$$-\frac{15}{8} \cdot 0 + \frac{13}{8} \cdot 0 = 0 \leq \frac{135}{8} \quad \text{ja}$$

grad 2 (2,0)

$$\vec{N} = 3 \cdot (2,0) \cdot (6,0)$$

$$\begin{cases} \frac{3}{8}x_4 - \frac{1}{8}x_5 = -\frac{3}{8} \\ -\frac{15}{8}x_4 + \frac{13}{8}x_5 = \frac{135}{8} \end{cases}$$

$$| \cdot 8 =$$

$$\Rightarrow \begin{cases} 3x_4 - x_5 = -3 & | \cdot 5 \\ -15x_4 + 13x_5 = 135 \end{cases}$$

$$\begin{cases} 15x_4 - 5x_5 = -15 \\ -15x_4 + 13x_5 = 135 \end{cases}$$

$$8x_5 = 120$$

$$x_5 = 15$$

$$3x_4 - 15 = -3$$

$$x_4 = 4$$

$$x_1 = -\frac{3}{8} - \frac{3}{8} \cdot 4 + \frac{1}{8} \cdot 15 = 0$$

$$x_2 = -9 - \frac{1}{8} \cdot 4 + \frac{3}{8} \cdot 15 = 4$$

$$x_3 = \frac{135}{8} + \frac{15}{8} \cdot 4 - \frac{13}{8} \cdot 15 = 0$$

$$Z = 2 \cdot 4 + 27 = 8 + 27 = 35$$

Antwort: $Z_{\max} = Z(0; 4; 0; 4; 15) = 35$